Chapter 13, Problem 17PP

Problem

In Example 13.17, if the eight 100-W bulbs are replaced by twelve 60-W bulbs and the kitchen range is replaced by a 4.5-kW air- conditioner, find: (a) the total power supplied, (b) the current *Ip* in the primary winding.

Example 13.17:

A distribution transformer is used to supply a household as in Fig. 13.68. The load consists of eight 100-W blbs, a 350-WTV, and a 15-kW kitchen range. If the secondary side of the transformer has 72 turns, calculate: (a) the number of turns of the primary winding, and (b) the current I_p in the primary winding.

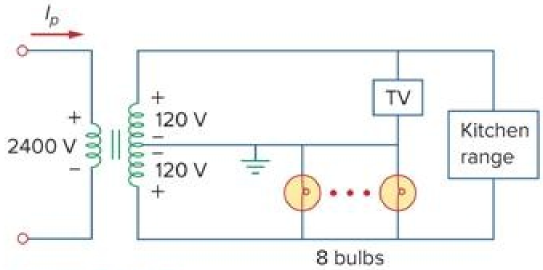


Figure 13.68
For Example 13.17.

Solution:

(a) The dot locations on the winding are not important, since we are only interested in the magnitudes of the variables involved. Since

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

we get

$$N_p = N_s \frac{V_p}{V_s} = 72 \frac{2,400}{240} = 720 \text{ turns}$$

(b) The total power absorbed by the load is

$$S = 8 \times 100 + 350 + 15,000 = 16.15 \text{ kW}$$

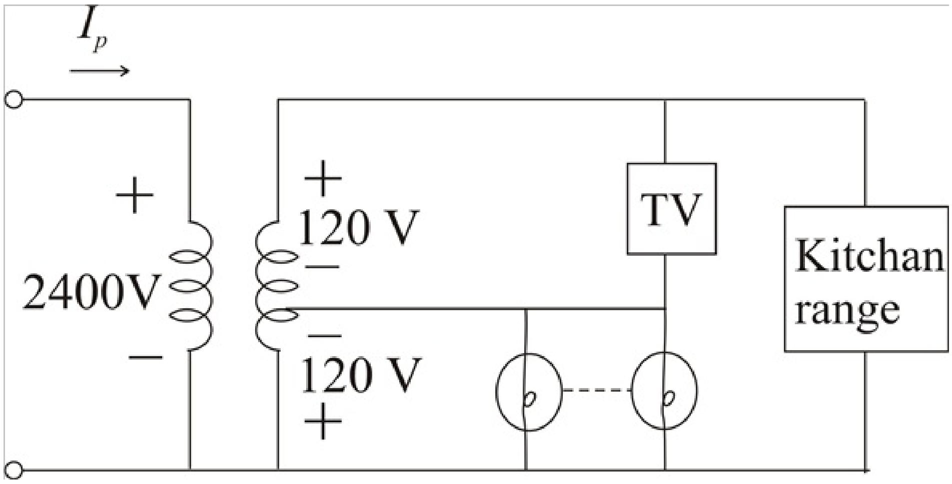
But $S = V_p I_p = V_s I_s$, so that

$$I_p = \frac{S}{V_p} = \frac{16,150}{2,400} = 6.729 \text{ A}$$

Step-by-step solution

Step 1 of 3

Given circuit,



Step 2 of 3

(a) The dot locations on the winding are not important, since we are only interested in the magnitudes of the variables involved. Since,

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

We get

$$N_p = N_s\left[\frac{V_p}{V_s}\right] \quad \text{.............. (1)}$$

Given that,

$$N_s = 72 \text{ turns}$$

$$60\text{W blubs } = 12$$

Kitchen range $= 4.5\text{kW}$

From (1)

$$N_p = 72\left[\frac{2400}{240}\right]$$

$$\therefore N_p = 720 \text{ turns}$$

The total power absorbed by the load is,

$$S = (12\times 60) + (350) + (4.5\times 10^3)$$

$$\therefore \boxed{S = 5.57\text{kW}}$$

Step 3 of 3

(b) We have

$$S = V_p I_p$$

$$I_p = \frac{S}{V_p}$$

$$I_p = \frac{5.5\times 10^3}{2400}$$

$$\therefore \quad \boxed{I_p = 2.321\text{A}}$$